

Variational Autoencoders

$$\max_{\theta} \log p_{\theta}(x) \rightarrow \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z))$$

$$\log p_{\theta}(x) = \log p_{\theta}(x) \cdot \int q_{\phi}(z|x) dz = \int q_{\phi}(z|x) \log p_{\theta}(x) dz = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x)]$$

$$= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{p_{\theta}(z|x)} \right]$$

$$= \mathbb{E}_{q_{\phi}(z|x)} \left[\log p_{\theta}(x|z) \right] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right]$$

$$= \mathbb{E}_{q_{\phi}(z|x)} \left[\log p_{\theta}(x|z) \right] - \text{KL}(q_{\phi}(z|x) \| p(z)) + \text{KL}(q_{\phi}(z|x) \| p_{\theta}(z|x))$$

$$\geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z)) \quad \therefore \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z))$$